

Week 4 — If/Else Decisions

Name: ____ Date: ____

A comparison is a **question**. The answer is *True* or *False*. *if* runs its indented block only when the answer is *True*.

Today's words

Word	What it means
boolean	a value with only two options: <i>True</i> or <i>False</i>
comparison	a question about values: <code>></code> <code><</code> <code>>=</code> <code><=</code> <code>==</code> <code>!=</code>
condition	the question an <code>if</code> checks before deciding
if / elif / else	if this... otherwise if this... otherwise. ONE branch runs
indent	the sideways space meaning "this line belongs to the if"

1 • Be the computer 🤖

Trace top-down and write the ONE line that prints:

```
temperature = 12
if temperature >= 28:
    print("melting")
elif temperature >= 15:
    print("perfect")
elif temperature >= 0:
    print("jacket")
else:
    print("frozen")
```

The screen shows: ____

2 • Closed or open circle? ○ ●

The ride rule is `height >= 132`. Circle the bouncer's answer for each rider:

Rider's height	Rides?
131	RIDES / TEACUPS
132	RIDES / TEACUPS
133	RIDES / TEACUPS

On a number line, `>= 132` is a **CLOSED / OPEN** (circle one) circle at 132. If the rule were `> 132`, the rider at exactly 132 would: ____

3 • Spot the bug 🐛

Each snippet has **one** mistake. Circle it.

- `if age = 13:`
 `print("teen")`
- `if age >= 13`
 `print("teen")`
- `if age >= 13:`
 `print("teen")` ← this line is supposed to belong to the if

🧠 Brain teaser (optional — take it home)

The hat with a hole in it. A student wrote this Sorting Hat, but ONE lucky number makes it misbehave. Which number, and what happens to it?

```
if lucky > 50:
    print("Team Waffle")
elif lucky < 50:
    print("Team Goat")
```

Then fix it: rewrite the rules (you're allowed to change `>` and `<` to anything from today) so that EVERY number from 1 to 100 lands on exactly one team. Bring your fix next week for a shout-out — bonus glory if you find *two* different ways to plug the hole.